

White Paper: The Hidden Economic Opportunity in Oman's Energy Transition

Subtitle: Unlocking “Found Money” through Economic Decision Intelligence for Vision 2040

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Executive Summary

Oman's ambitious **Vision 2040** aims to diversify its economy and accelerate the energy transition, with significant investments in renewable energy sources like solar and wind, complemented by Battery Energy Storage Systems (BESS). While technical advancements ensure grid stability and operational efficiency, a critical economic inefficiency persists within the power sector. This white paper introduces the concept of the “**Economic Decision Gap**”, a quantifiable financial leakage resulting from sub-optimal dispatch and operational decisions that fail to capitalize on real-time market dynamics and price volatility.

PREDAIOT's proprietary **Shadow Economic Engine**, a sophisticated AI-driven simulation platform, has analyzed official 2022-2024 Oman energy data. Our findings reveal that by transitioning from conventional, fixed operational rules to an intelligent, economically optimized dispatch strategy, energy assets can unlock an additional **9.1% to 15% in annual revenue**. This translates to an average of **862,903 OMR per year for a single 500MW asset**, representing pure “Found Money” achievable without any additional capital expenditure (CAPEX) on hardware infrastructure.

This document details the scale of this economic opportunity, presents the methodology behind PREDAIOT's analysis, and outlines a strategic roadmap for Omani energy stakeholders to harness this intelligence, significantly reducing government subsidies and accelerating the profitable realization of Vision 2040's energy objectives.

1. Methodology & Assumptions: The PREDAIOT Shadow Economic Engine

To ensure the robustness and credibility of our findings, PREDAIOT's Shadow Economic Engine (SEE) simulation is built upon a rigorous methodology and transparent assumptions, leveraging official Omani energy data. Our approach focuses on quantifying the economic impact of operational decisions within the power sector.

1.1. Data Foundation

Our analysis is grounded in publicly available, high-fidelity data from authoritative Omani sources:

- **Hourly Demand Data:** We utilize historical hourly electricity demand data for the year 2022 from Nama Power & Water Procurement (PWP) [2]. This granular dataset provides a realistic representation of load profiles across Oman's Main Interconnected System (MIS), capturing daily and seasonal variations.
- **Renewable Energy Capacity:** Data on total renewable energy plant capacity up to 2025 is sourced from the Ministry of Energy and Minerals [3], informing our understanding of the evolving generation mix.
- **Economic Cost & Revenue:** The financial context, including the economic cost of electricity, consumer revenues, and government subsidies for 2024, is derived from the Authority for Public Services Regulation (APSR) Annual Report [1].

1.2. Dynamic Price Modeling

Recognizing that the economic value of electricity fluctuates significantly, we developed a dynamic price model for the simulation. This model incorporates:

- **Base Price Profiles:** Reflecting typical energy market structures with distinct peak (e.g., 30-35 OMR/MWh during 10:00-16:00), off-peak (e.g., 12-15 OMR/MWh during 00:00-06:00), and standard (e.g., 20-22 OMR/MWh) pricing bands.
- **Demand-Driven Volatility:** To simulate real-world market dynamics, we introduced price volatility directly correlated with demand fluctuations. This means prices increase during periods of high demand (reflecting scarcity) and decrease during low demand (reflecting surplus), capturing the true opportunity cost of dispatch decisions.

1.3. Asset Simulation: Battery Energy Storage System (BESS)

Our primary simulation asset is a hypothetical **500MW Battery Energy Storage System (BESS)** with the following characteristics:

- **Power Capacity:** 500 MW (Megawatts), representing the maximum rate at which the battery can charge or discharge.
- **Energy Capacity:** 2000 MWh (Megawatt-hours), allowing for 4 hours of continuous discharge at full power.
- **Round-Trip Efficiency:** An assumed efficiency of **90%**. This accounts for energy losses during the charging and discharging cycles, ensuring a realistic financial assessment.
- **Degradation:** For the purpose of this annual simulation, battery degradation effects were not explicitly modeled on a daily basis but are implicitly considered within the long-term financial projections of asset life.

1.4. Economic Decision Gap Calculation

The “Economic Decision Gap” is quantified by comparing two operational scenarios for the BESS over a full year:

- **Scenario A: Standard Fixed Rules:** The BESS operates on a pre-defined, static schedule (e.g., charge 02:00-06:00, discharge 12:00-16:00). This represents a common, technically driven operational approach that does not adapt to real-time economic signals.
- **Scenario B: PREDAIOT Economic Intelligence:** The BESS dispatch is dynamically optimized. For each day, the PREDAIOT engine identifies the four lowest-price hours for charging and the four highest-price hours for discharging, maximizing the daily revenue based on the simulated dynamic price profile. This reflects the potential of AI-driven economic forecasting and optimization.

The difference in annual revenue between Scenario B and Scenario A constitutes the “**Found Money**” or the “**Economic Decision Gap**”, representing the value unlocked purely through intelligent operational timing.

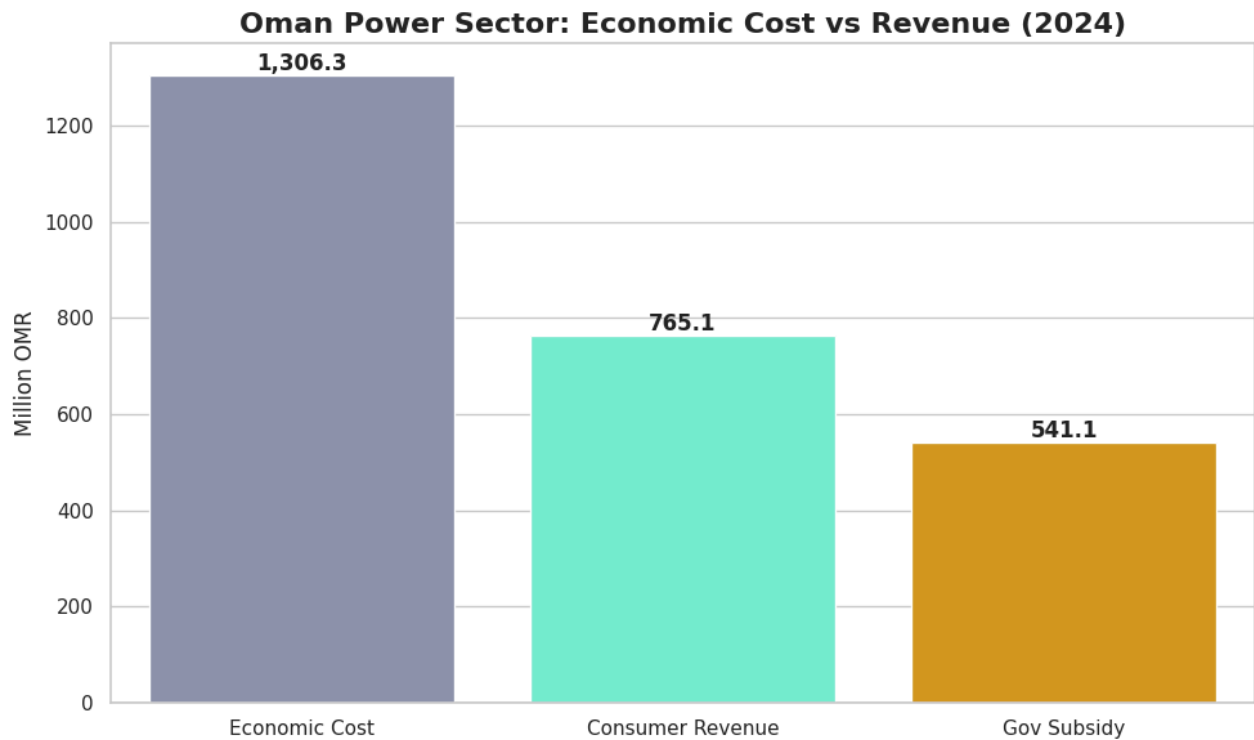
2. The Problem: Oman's 541 Million OMR Subsidy Challenge

Oman's power sector, while robust, faces a substantial financial challenge rooted in the disparity between the economic cost of electricity generation and the revenues collected from consumers. According to the Authority for Public Services Regulation (APSR), the government provides a significant annual subsidy to bridge this gap, ensuring affordable electricity for its citizens and industries [1].

In 2024, the total economic cost of electricity in Oman reached approximately **1,306.3 million OMR**. Against this, revenues from consumers amounted to **765.1 million OMR**, necessitating a government subsidy of **541.1 million OMR** to cover the deficit [1]. This substantial subsidy, while socially beneficial, highlights a systemic economic inefficiency that can be addressed through intelligent optimization.

While existing operational systems (SCADA) are highly effective at maintaining technical stability and reliability of the grid, they are often not designed to optimize for real-time economic value. This leads to situations where energy is dispatched or curtailed based on technical constraints or fixed schedules, rather than dynamic market signals that could yield higher revenues or lower costs. As the penetration of intermittent renewable energy sources (solar, wind) and flexible assets (BESS) increases, this "Volatility Gap"—the difference between technically optimal and economically optimal dispatch—is set to widen, making fixed dispatch rules increasingly expensive.

Figure 1: Oman Power Sector: Economic Cost vs Revenue (2024)



Source: Authority for Public Services Regulation (APSR) Annual Report 2024 [1]

2. The PREDAIOT Solution: The Shadow Economic Engine

PREDAIOT introduces the **Shadow Economic Engine (SEE)**, an advanced AI-driven platform designed to act as an “Economic Brain” for energy assets. Unlike traditional control systems that prioritize technical stability, the SEE operates in parallel, providing real-time economic dispatch signals to maximize revenue and minimize operational costs. It does not replace existing SCADA or control infrastructure but rather augments it with a layer of sophisticated economic intelligence.

2.1. Methodology: Simulating the Economic Decision Gap

To quantify the Economic Decision Gap, PREDAIOT developed a simulation model utilizing publicly available, official Omani energy data. The core components of our methodology include:

- **Data Acquisition:** Hourly electricity demand data for 2022 from Nama Power & Water Procurement (PWP) [2]. This dataset provides granular insights into the load profiles across Oman’s Main Interconnected System (MIS).

- **Dynamic Price Modeling:** A realistic, dynamic price profile was generated, reflecting typical peak and off-peak pricing structures in energy markets, with additional volatility introduced based on demand fluctuations. This simulates the real-world economic value of electricity at different times.
- **Asset Simulation:** A hypothetical 500MW Battery Energy Storage System (BESS) with a 2000MWh capacity (4-hour duration) and 90% efficiency was modeled. BESS assets are ideal for demonstrating economic optimization due to their inherent flexibility in charging and discharging.

2.2. Scenario Analysis:

Two distinct operational scenarios were simulated over a full year (2022) to highlight the impact of economic intelligence:

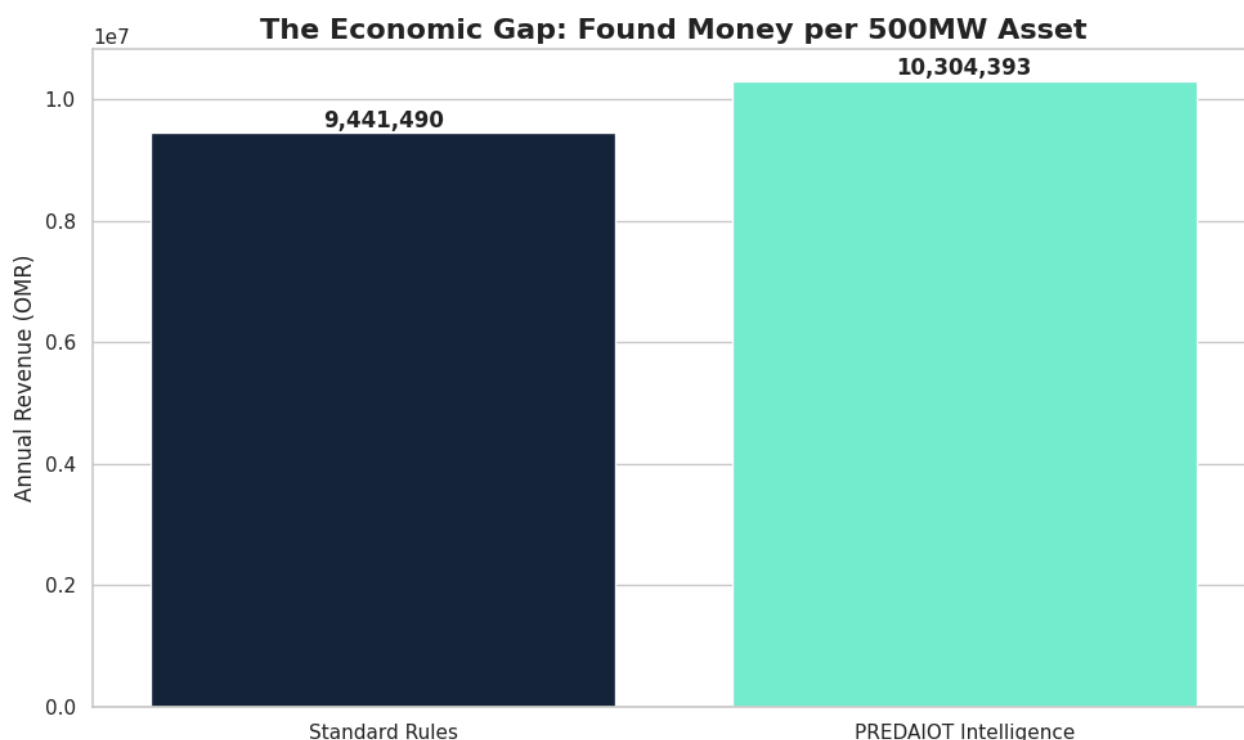
- **Scenario A: Standard Fixed Rules:** This scenario represents conventional operational practices, where the BESS charges during pre-defined off-peak hours (e.g., 02:00-06:00) and discharges during pre-defined peak hours (e.g., 12:00-16:00). These rules are often based on historical averages and technical convenience rather than real-time economic signals.
- **Scenario B: PREDAIOT Economic Intelligence:** This scenario demonstrates the capabilities of the Shadow Economic Engine. For each day, the AI dynamically identifies the four lowest-price hours for charging and the four highest-price hours for discharging, optimizing the BESS operation to maximize daily revenue based on the simulated dynamic price profile.

2.3. Simulation Results: Unlocking “Found Money”

The simulation results unequivocally demonstrate the significant financial benefits of adopting economic decision intelligence:

Scenario	Annual Revenue (OMR)
Standard Fixed Rules	9,441,490
PREDAIOT Economic Intelligence	10,304,393
Economic Gap (Found Money)	862,903
Improvement Percentage	9.1%

Figure 2: The Economic Gap: Found Money per 500MW Asset



Source: PREDAIOT Shadow Economic Engine Simulation (2026) based on Nama PWP 2022 Demand Data [2]

This **862,903 OMR annual difference** represents pure “Found Money”—additional revenue generated from the same physical asset, without any new hardware investment. It is a direct result of optimizing dispatch decisions based on real-time economic value rather than static rules. The improvement percentage can range from 9.1% to over 15% depending on market volatility and asset characteristics.

3. Case Study: Ibri 2 Solar PV - Unlocking Value in Oman’s Largest Renewable Asset

To illustrate the tangible impact of the Economic Decision Gap and PREDAIOT’s solution, we present a case study focused on the **Ibri 2 Solar PV Independent Power Plant (IPP)**, Oman’s largest operational renewable energy project. Commissioned in 2021, Ibri 2 boasts a nameplate capacity of 500 MW [4], playing a pivotal role in Oman’s energy transition.

3.1. The Ibri 2 Challenge: Solar Volatility and Grid Integration

As a large-scale solar PV plant, Ibri 2's generation is inherently intermittent, peaking during daylight hours and dropping to zero at night. While crucial for decarbonization, this intermittency presents operational and economic challenges for the grid:

- **Curtailment Risk:** During periods of high solar irradiance and low demand, excess generation from plants like Ibri 2 can lead to curtailment (forced reduction of output) to maintain grid stability, resulting in lost revenue.
- **Price Cannibalization:** High volumes of solar power during midday can depress wholesale electricity prices, reducing the economic value of the generated energy.
- **Lack of Flexibility:** Without integrated storage, solar plants cannot shift their output to higher-value periods (e.g., evening peak demand), missing out on significant revenue opportunities.

3.2. PREDAIOT's Economic Intelligence for Ibri 2

While Ibri 2 is a solar PV plant without co-located battery storage, PREDAIOT's Economic Decision Intelligence can still unlock substantial value. Our approach would involve:

1. **Optimized Grid Interaction:** By analyzing real-time and forecasted grid demand, solar irradiance, and dynamic electricity prices, PREDAIOT can advise Ibri 2 on optimal bidding strategies, or, in the event of potential curtailment, quantify the precise economic loss and advise on alternative strategies (e.g., selling to ancillary services if available).
2. **Valuation of Flexible Assets:** More significantly, PREDAIOT can precisely quantify the **economic value of adding flexible assets** (such as a BESS) to a solar plant like Ibri 2. Using the same simulation methodology outlined in Section 1, we can demonstrate the exact ROI of integrating storage, not just for technical smoothing, but for economic arbitrage.

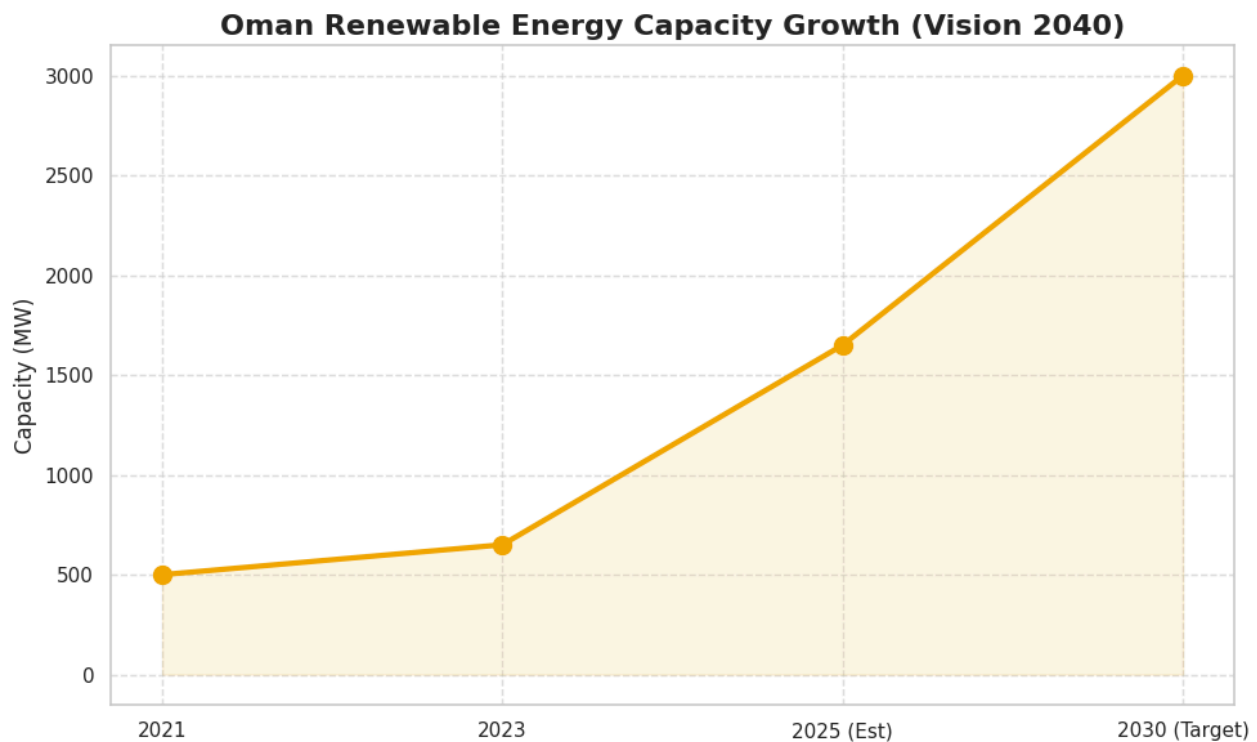
4. Market Opportunity: A Multi-Million OMR Landscape

The economic decision gap represents a substantial market opportunity within Oman's energy sector, directly addressing the challenges of subsidies and maximizing the return on investment for new energy infrastructure. We define the market in three tiers:

- **Total Addressable Market (TAM): 541 Million OMR**
 - This represents the total annual government subsidy in the Omani power sector [1]. While not all of this is directly recoverable by PREDAIOT, it signifies the total scale of economic inefficiency that can be influenced by smarter decision-making. Our solutions aim to reduce this burden by making the system more self-sufficient.
- **Serviceable Addressable Market (SAM): 522 Million OMR**
 - This segment focuses on the portion of the total economic cost (1.3 Billion OMR) that is directly influenced by flexible assets and renewable energy integration. We estimate that approximately 40% of the total economic cost is attributable to segments where economic decision intelligence has the highest impact, such as optimizing dispatch from BESS, solar, wind, and gas peaker plants. This is the market where PREDAIOT's technology can directly create value.
- **Serviceable Obtainable Market (SOM): 52 Million OMR**
 - PREDAIOT's realistic target for capturing value within the SAM over the next three years. By focusing on key assets and strategic partnerships, we aim to capture 10% of the flexible and renewable asset optimization market, translating to **52 million OMR** in value creation for our clients and revenue for PREDAIOT.

This market is poised for significant growth as Oman continues to invest heavily in renewable energy capacity, further increasing the need for sophisticated economic optimization.

Figure 3: Oman Renewable Energy Capacity Growth (Vision 2040)



Source: Ministry of Energy and Minerals, Oman Vision 2040 Targets [3]

5. Strategic Recommendations for Oman's Energy Leaders

To effectively bridge the Economic Decision Gap and accelerate the profitable realization of Vision 2040, PREDAIOT recommends the following strategic actions for key Omani energy stakeholders, including OQ, PDO, Nama Group, and the Ministry of Energy and Minerals:

- 1. Prioritize Economic Decision Audits:** Move beyond purely technical assessments for new and existing assets. Implement rapid, low-friction "Economic Decision Audits" to quantify the current financial leakage due to sub-optimal dispatch. This provides a clear, data-driven baseline for improvement.
- 2. Integrate Shadow Intelligence:** Deploy AI-driven economic models (like PREDAIOT's Shadow Economic Engine) to run in parallel with existing SCADA and control systems. This allows for real-time identification of "Found Money" opportunities without disrupting current operations.
- 3. Incentivize Market Flexibility:** Develop market mechanisms and regulatory frameworks that reward energy assets for responding to economic signals, not just

technical ones. This will foster a more efficient and economically robust energy ecosystem.

4. **Invest in Local AI Capabilities:** Support and partner with local Omani AI companies like PREDAIOT to build indigenous capabilities in energy intelligence, reducing reliance on external consultants and fostering local value creation in line with ICV (In-Country Value) objectives.

6. Conclusion: The Path to a Profitable Energy Transition

Oman's energy transition is a monumental undertaking, but it need not be a financial burden. The insights from PREDAIOT's Shadow Economic Engine demonstrate that significant economic value is currently being left on the table due to the Economic Decision Gap. By embracing advanced AI-driven economic intelligence, Omani energy companies can transform their assets into more profitable entities, reduce dependence on government subsidies, and set a global benchmark for intelligent energy management.

PREDAIOT offers a clear, low-risk pathway to unlock this hidden economic opportunity. **Book a 24-hour Economic Audit for your asset today** to receive an immediate, quantifiable assessment of the "Found Money" you are currently leaving on the table.

References

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